

IT Security 2

Unit 11 - Blocking UNKNOWN THREATS with Advanced Wildfire

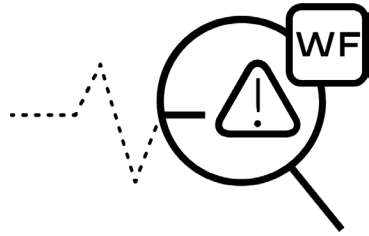
Instructor: Brian Krause

Announcements

Blocking UNKNOWN THREATS with Advanced Wildfire

Some slides were adapted from Palo-Alto for use in this class. Content is a mesh between vendor provided information and instructor comments.

BLOCKING UNKNOWN THREATS WITH ADVANCED WILDFIRE



*IF I AM PUTTING MYSELF OUT THERE AND TAKING SOME OF
THESE RISKS, THEN I WANT TO DO IT PROPERLY*

- WildFire[®] concepts
- Configure and manage WildFire
- WildFire reporting

Learning Objectives

After you complete this module, you should be able to:

- Describe WildFire purposes and operation
- Describe WildFire license and deployment choices
- Configure and update WildFire
- View WildFire reports and logs



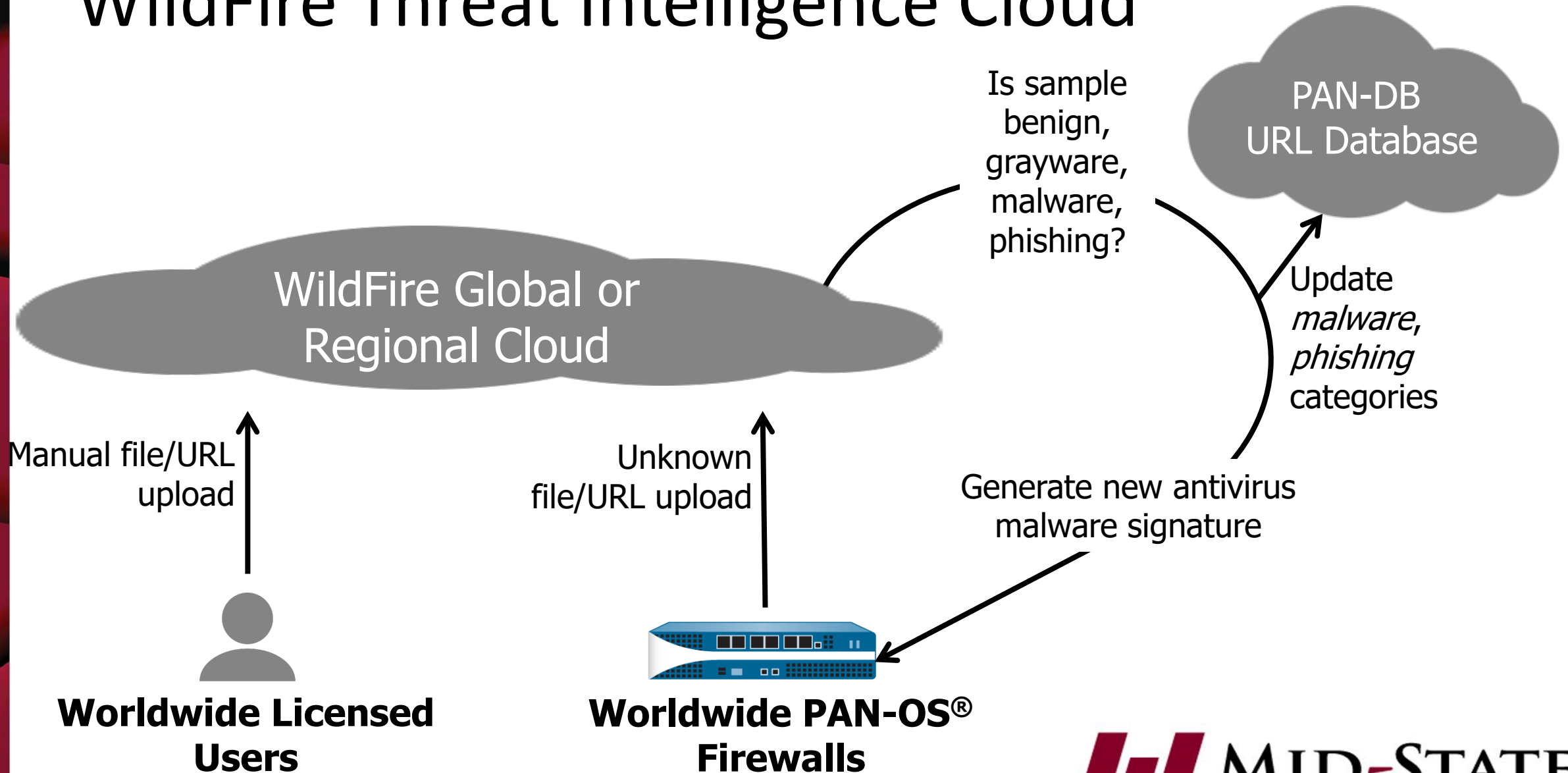


WildFire concepts

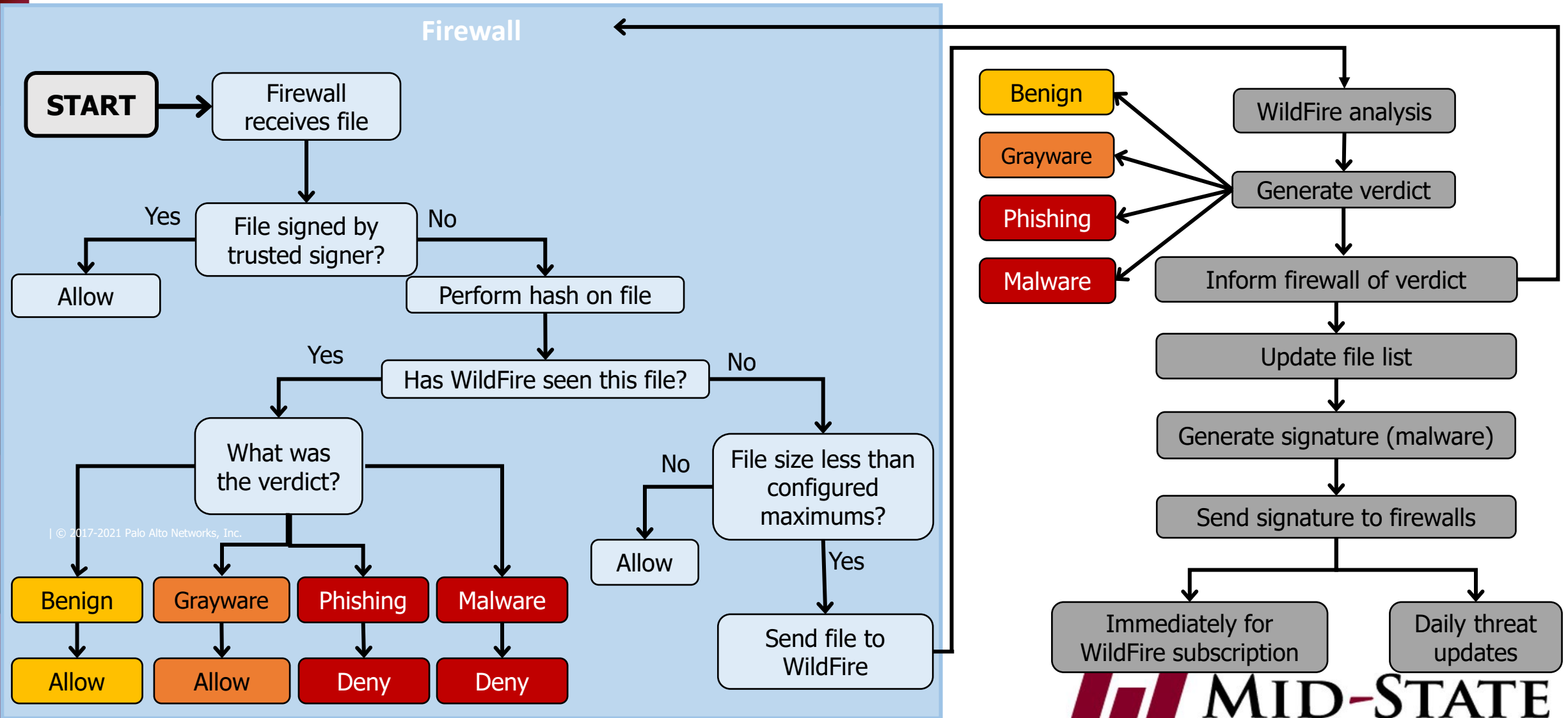
Configure and manage WildFire

WildFire reporting

WildFire Threat Intelligence Cloud



WildFire Operation Overview

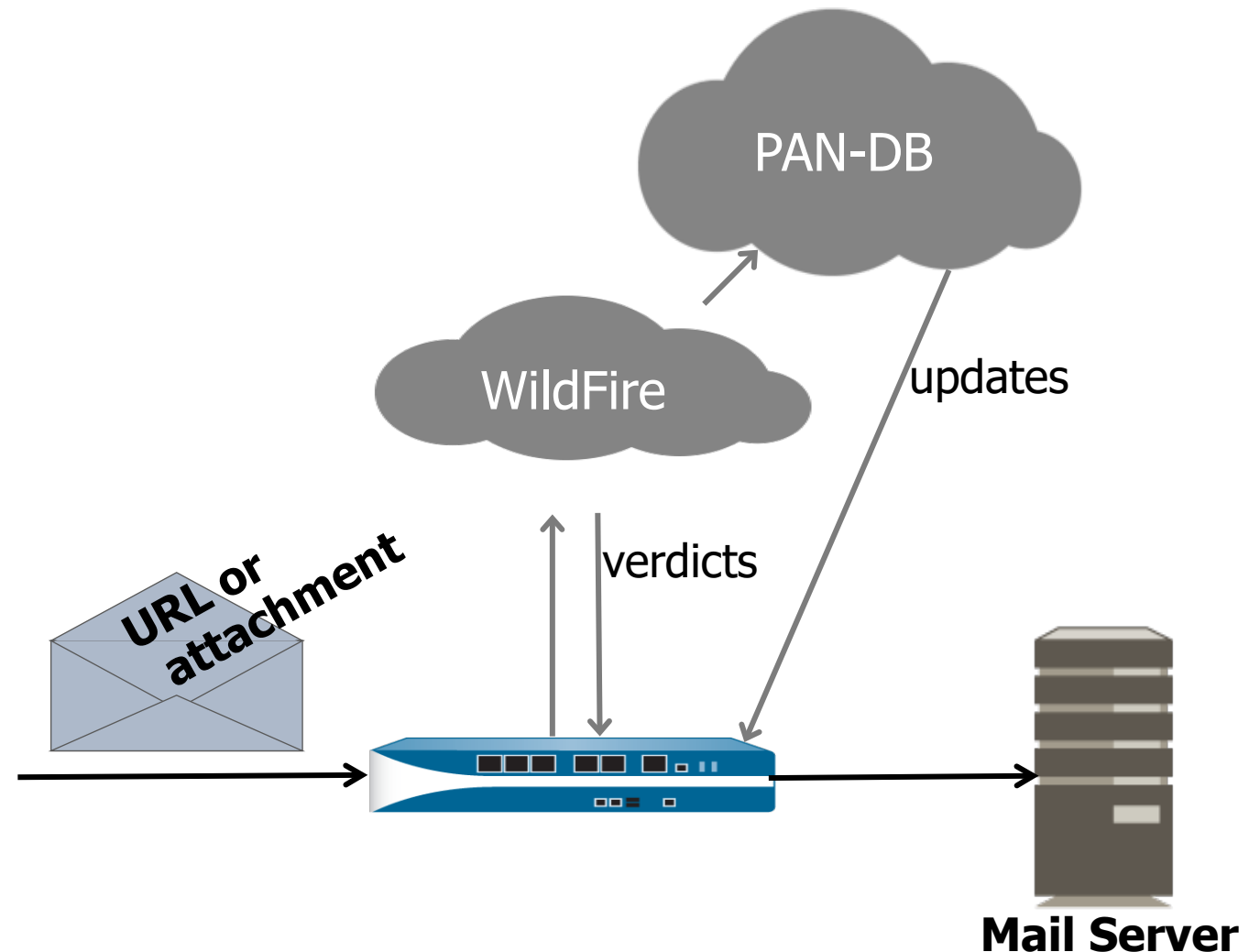


WildFire Verdict Descriptions

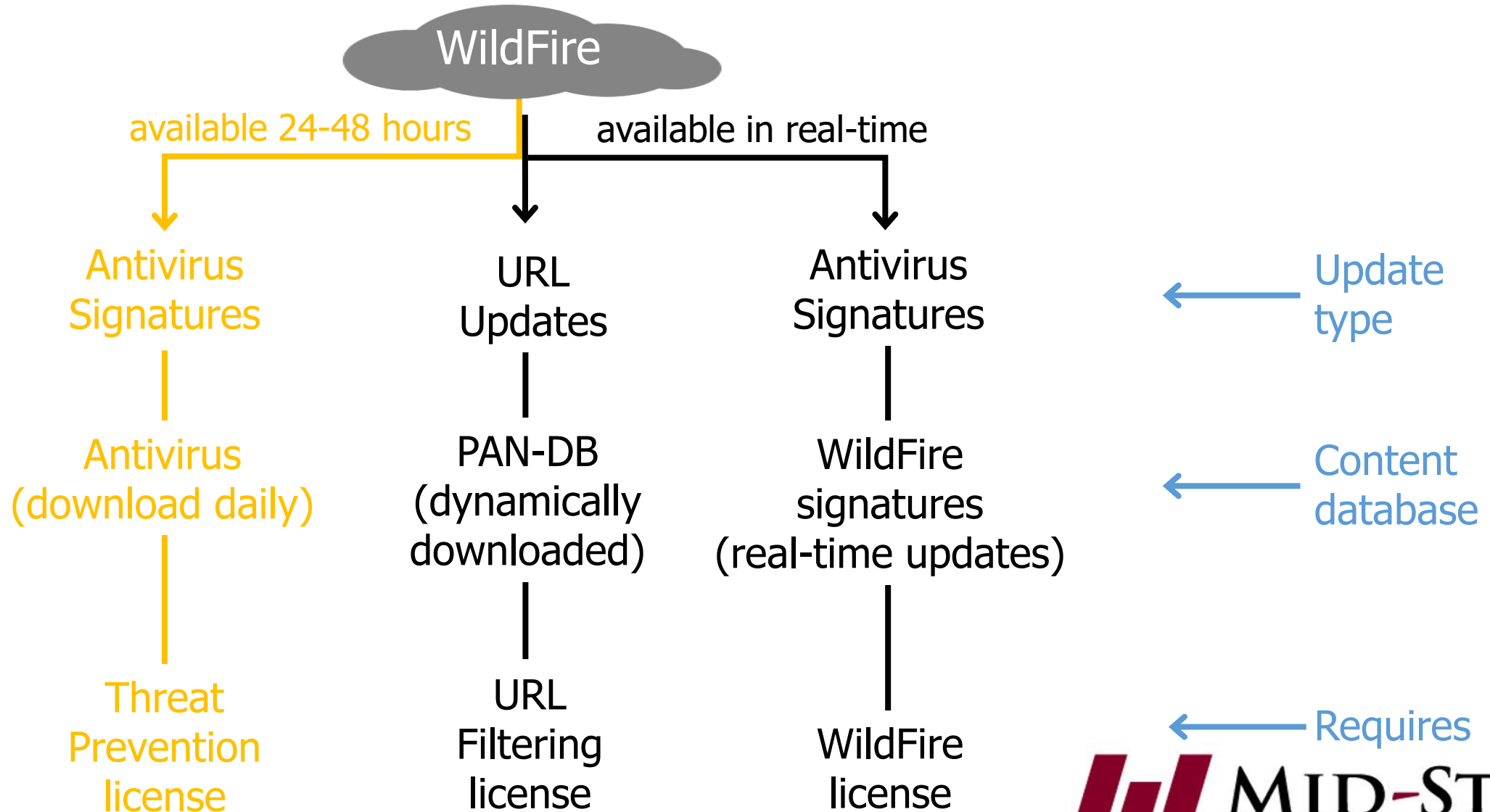
Verdict	Description
Benign	• Safe and does not exhibit malicious behavior.
Grayware	• No security threat but might display obtrusive behavior. • Examples include adware, spyware, and browser helper objects (BHOs).
Malicious	• Malicious in nature and intent and can pose a security threat. • Examples include viruses, worms, trojans, remote access tools (RATs), rootkits, and botnets.
Phishing	• An attempt to trick users into revealing their login information. • Based on properties and behaviors the website displays.

WildFire Protects Email

- Email with attachments or URL links is sent to WildFire for analysis.
- If an attachment or link is malicious, WildFire can:
 - Create and download new antivirus signatures to the firewall
 - Update the PAN-DB database with malicious URLs
- The firewall uses new information to protect the network.



Content Packages and WildFire Updates



Basic and Licensed Functionality

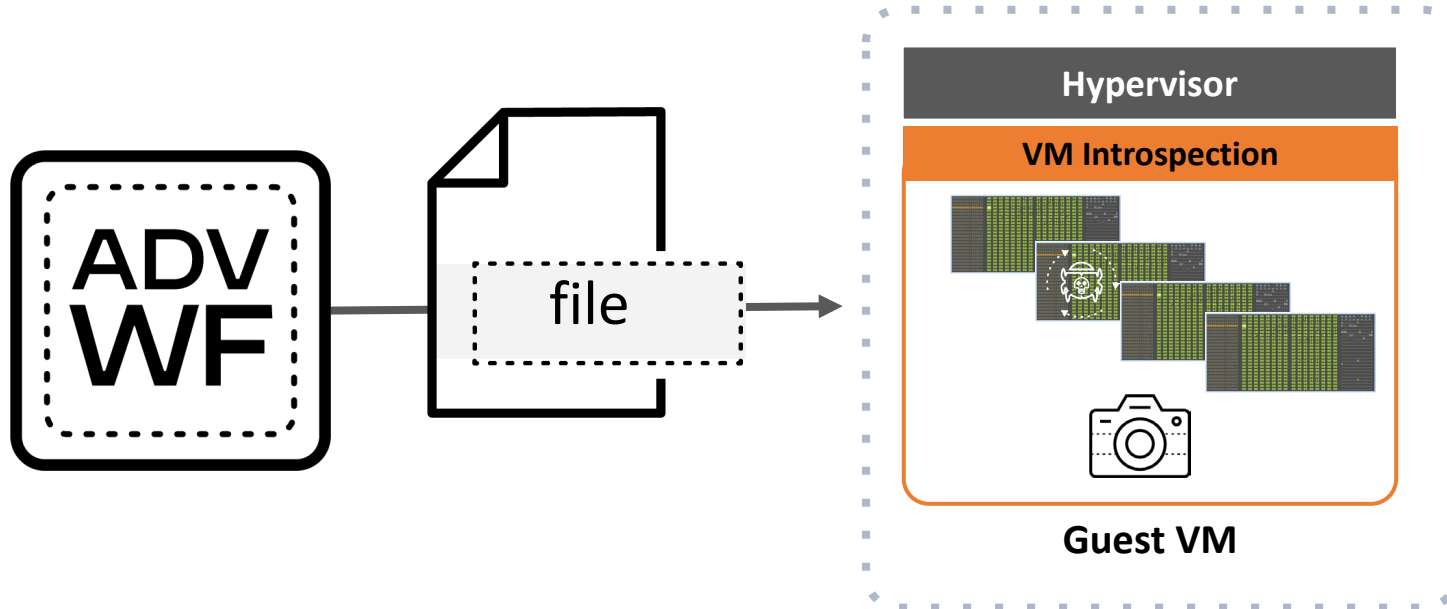
Standard subscription service:

- Analysis available in Windows XP, 7, 10, macOS, Android, Linux, and bare metal
- Windows PE file analysis:
 - EXE, DLL, FON, SCR, others
- Antivirus signatures delivered via daily dynamic content updates (requires Threat Prevention license)
- Automatic file submission

WildFire subscription service:

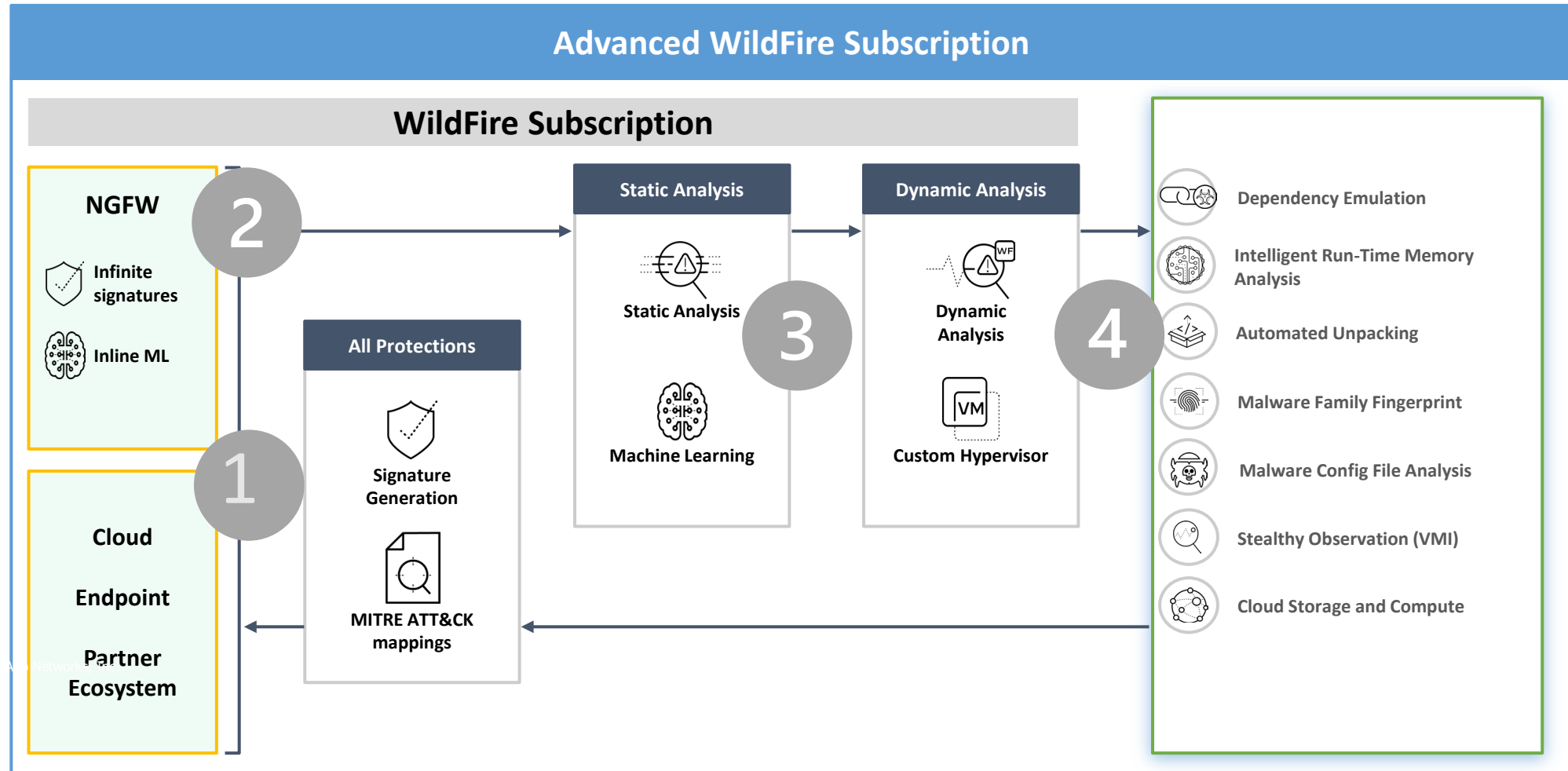
- Standard subscription features
- Additional file type analysis:
 - Microsoft Office, PDF, JAR, CLASS, SWF, SWC, RAR, 7-Zip, Linux ELF, APK, Mach-O, DMG, PKG, JS, VBS, PS1
- WildFire signature updates in real time
- Manual file submission via API
- WildFire private cloud appliance:
 - WF-500/WF-600

Advanced Wildfire License



Advanced WildFire	
	Dependency Emulation
	Intelligent Memory Analysis
	Automated Unpacking
	Malware Family Fingerprint
	Malware Config File Analysis
	Stealthy Observation (VMI)
	Cloud Storage and Compute
Includes the leading WildFire Malware Analysis Capabilities	

Advanced WildFire Subscription Service

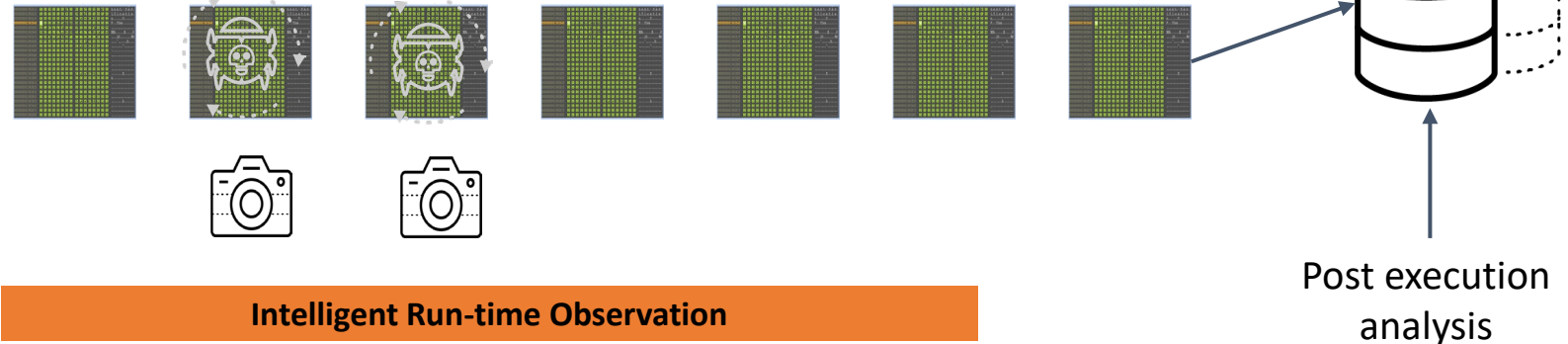


Intelligent Run-Time Memory Analysis

Advanced WildFire

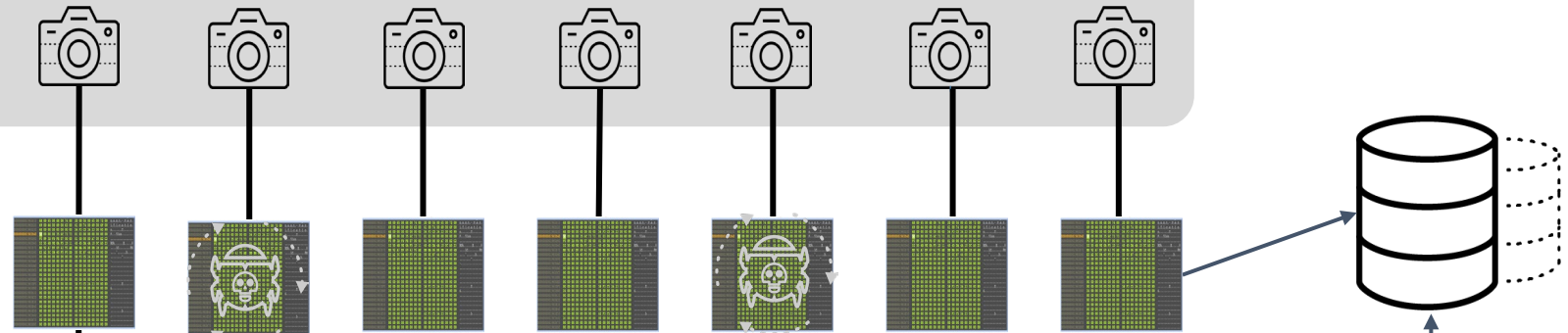
Intelligent Memory Capture

Malicious activity is captured during runtime using fewer resources.



Continuous Memory Capture

Consumes memory and processing



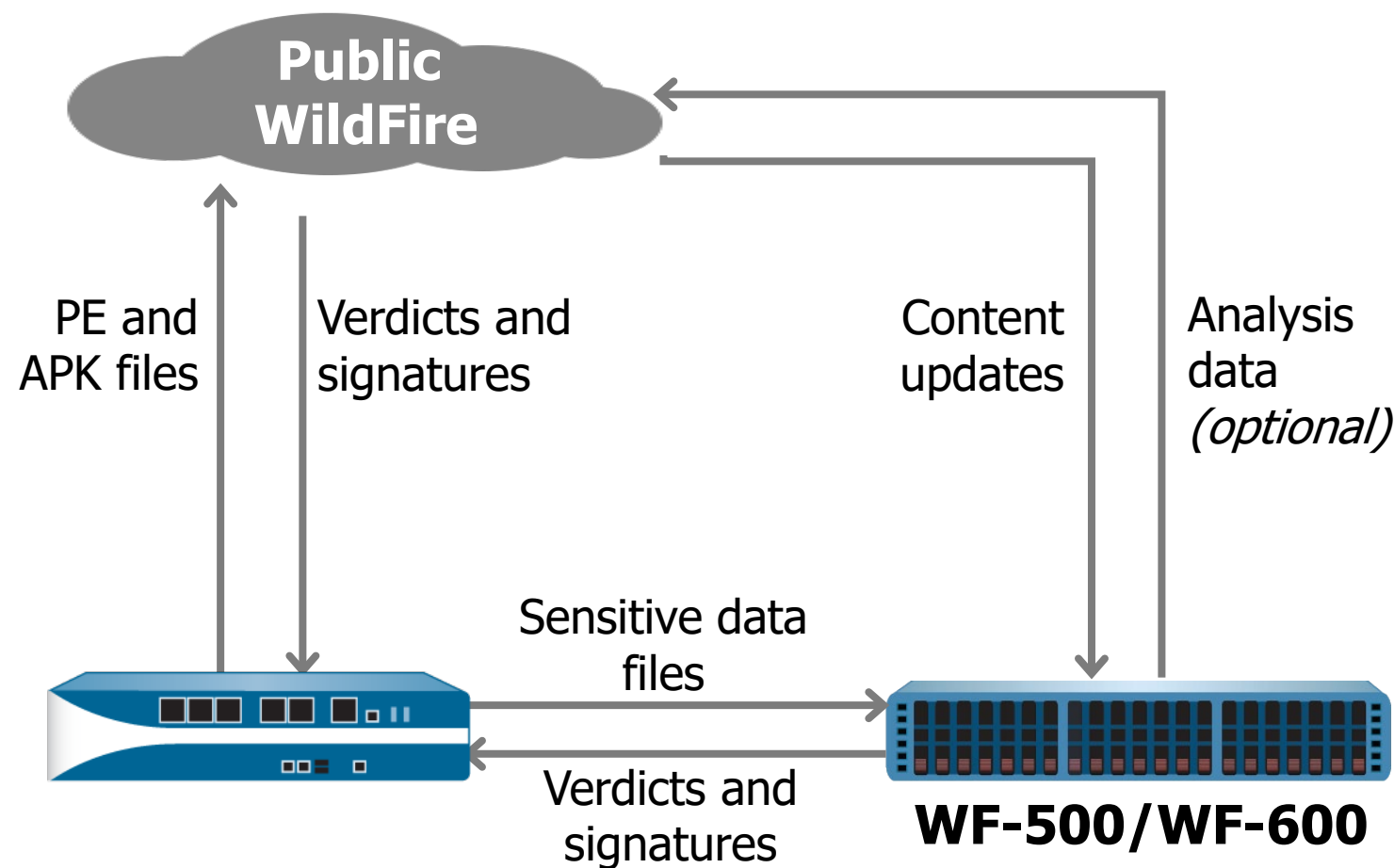
Intermittent Memory Capture

May miss malware execution



Hybrid Cloud Example

- Combines public and private cloud
- PE and APK files to public cloud?
- Sensitive data files to private cloud?



WildFire concepts



Configure and manage WildFire

WildFire reporting

Configure WildFire Settings

Device > Setup > WildFire

General Settings

WildFire Public Cloud wildfire.paloaltonetworks.com

WildFire Private Cloud

☐ Use Proxy Settings for Private Cloud

File Size Limits

FILE TYPE	SIZE LIMIT
pe (MB)	16 (default)
apk (MB)	10 (default)
pdf (KB)	3072 (default)
ms-office (KB)	16384 (default)
jar (MB)	5 (default)
flash (MB)	5 (default)
MacOSX (MB)	10 (default)
archive (MB)	50 (default)
linux (MB)	50 (default)
script (KB)	20 (default)

☒ Report Benign Files

☒ Report Grayware Files

Configure connection to WildFire cloud(s).

Files that exceed size are not forwarded to WildFire.

Benign and grayware files appear in the WildFire Submissions log.

Note: Decrypted content is not forwarded to WildFire by default.

Submission Settings

Device > Setup > WildFire

Session Information Settings

☒

Source IP

☒

Source port

☒

Destination IP

☒

Destination port

☒

Virtual System

☒

Application

☒

User

☒

URL

☒

File name

☒

Email sender

☒

Email recipient

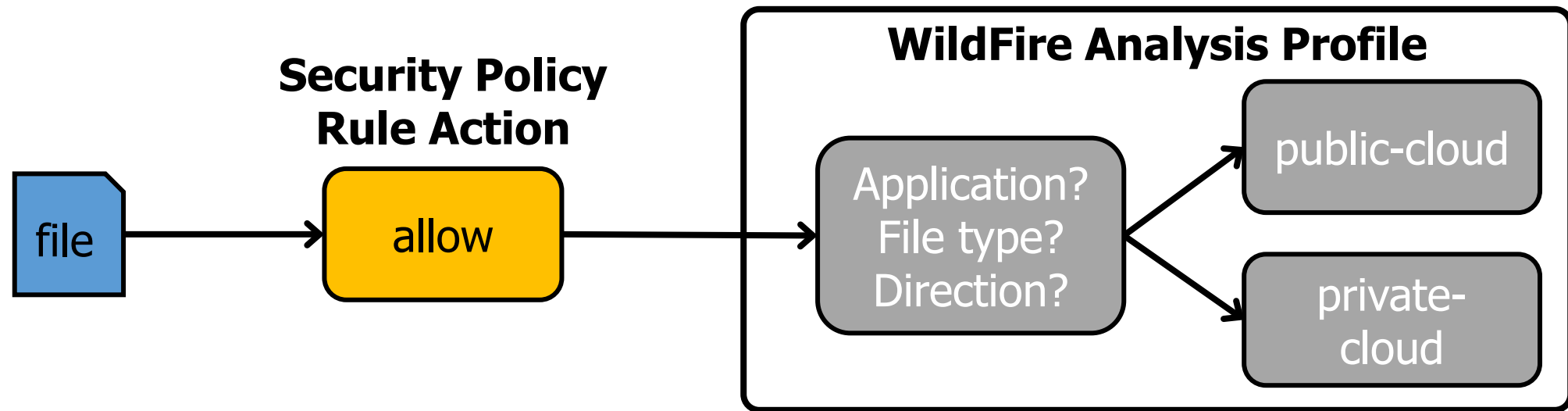
☒

Email subject

Define session information types reported to WildFire (and therefore available in WildFire Submissions log).

WildFire Analysis

Profile implements additional security checks on files in allowed traffic.



For a Tech Doc about this topic, log into Live and search for "WildFire Best Practices"

WildFire Analysis Profile

Objects > Security Profiles > WildFire Analysis

☐	NAME	RULE NAME	APPLICATIONS	FILE TYPES	DIRECTION	ANALYSIS
☐	default	default	any	any	both	public-cloud
+ Add - Delete						

PAN-OS built-in profile

Default rule sends all unknown files allowed by the Security policy rule to the WildFire public cloud.

- To create custom profiles:
 - Clone the read-only *default* profile and edit the clone, or
 - Create a new profile

For a Tech Doc about this topic, log into Live and search for "Objects > Security Profiles > WildFire Analysis"

Creating a WildFire Analysis Profile

Objects > Security Profiles > WildFire Analysis > Add

WildFire Analysis Profile

Name

Corp-WF

Description

Wildfire profile for Corp security rules

4 items

☐	NAME	APPLICATIONS	FILE TYPES	DIRECTION	ANALYSIS	
☐	apk	any	apk	both	public-cloud	
☐	Safe for Public	any	jar pe	both	public-cloud	
☐	Check Email	any	email-link	both	public-cloud	
☐	Not Safe for Public	any	ms-office	both	private-cloud	

Add

Delete

Configure Real-Time WildFire Analysis

Objects > Security Profiles > AntiVirus

Antivirus Profile

Name: WildFire-Real-Time-Analysis

Description: WildFire Inline Machine Learning Real-Time Analysis Profile

Action | Signature Exceptions: **WildFire Inline ML**

Available Models

MODEL	DESCRIPTION	ACTION SETTING
Windows Executables	Machine Learning engine to dynamically identify malicious PE files	disable (for all protocols)
PowerShell Script 1	Machine Learning engine to dynamically detect malicious PowerShell scripts with known length	enable (inherit per-protocol actions)
PowerShell Script 2	Machine Learning engine to dynamically detect malicious PowerShell scripts without known	disable (for all protocols)

File Exceptions

PARTIAL HASH	FILENAME
--------------	----------

(Optionally) Add file exceptions to exclude specific files.

Classification engines define each file type to apply to a policy.

Define policy action for each classification engine.

enable (inherit per-protocol actions)
alert-only (override more strict actions to alert)
disable (for all protocols)

Attach WildFire Analysis Profiles to Security Rules

Policies > Security > Add

Security Policy Rule

General | Source | Destination | Application | Service/URL Category | **Actions** | Usage

Action Setting

Action: Allow

☐ Send ICMP Unreachable

Profile Setting

Profile Type: Profiles

Antivirus: None

Vulnerability Protection: None

Anti-Spyware: None

URL Filtering: None

File Blocking: None

Data Filtering: None

WildFire Analysis: Corporate-WildFire-Profile

Log Setting

☐ Log at Session Start

☒ Log at Session End

Profile Setting (Callout)

Profile Type: Group

Group Profile: Corp-Profiles-Group

☐ Disable Server Response Inspection

- Add WildFire Analysis Profile to Security policy rule, or
- Add WildFire Analysis Profile to Group Profile and add group to Security policy rule.

WildFire Update Schedule

- Schedule poll period for WildFire antivirus signature updates:
 - Requires a WildFire license
 - Without a license, WildFire antivirus signatures still are added to the daily Antivirus content package.

Device > Dynamic Updates

The screenshot shows the 'Dynamic Updates' configuration for WildFire. At the top, a summary bar indicates 'Last checked: 2021/03/19 18:40:00 UTC' and 'Schedule: Real-time'. Below this is a table of update schedules. A modal window titled 'WildFire Update Schedule' is open, showing the configuration for a specific update. The 'Recurrence' dropdown is set to 'Every Minute', and the 'Action' dropdown is set to 'download-and-install'. A checkbox for 'Synchronize content with HA peer after download/install' is present and unchecked. Two callout boxes provide the full list of options for the 'Recurrence' and 'Action' dropdowns.

Device ID	Device Name	Version	Last Checked	Notes
467667-470604	panupv3-all-wildfire-467667-470604	PAN-OS 10.0 And Later	2020/07/01 17:27:16 UTC	
472271-475208				e Notes
542662-545681				e Notes

WildFire Update Schedule

Recurrence: Every Minute

Action: download-and-install

☐ Synchronize content with HA peer after download/install

None (Manual)

Real-time

Every Minute

Every 15 Minutes

Every 30 Minutes

Every Hour

None

download-only

download-and-install

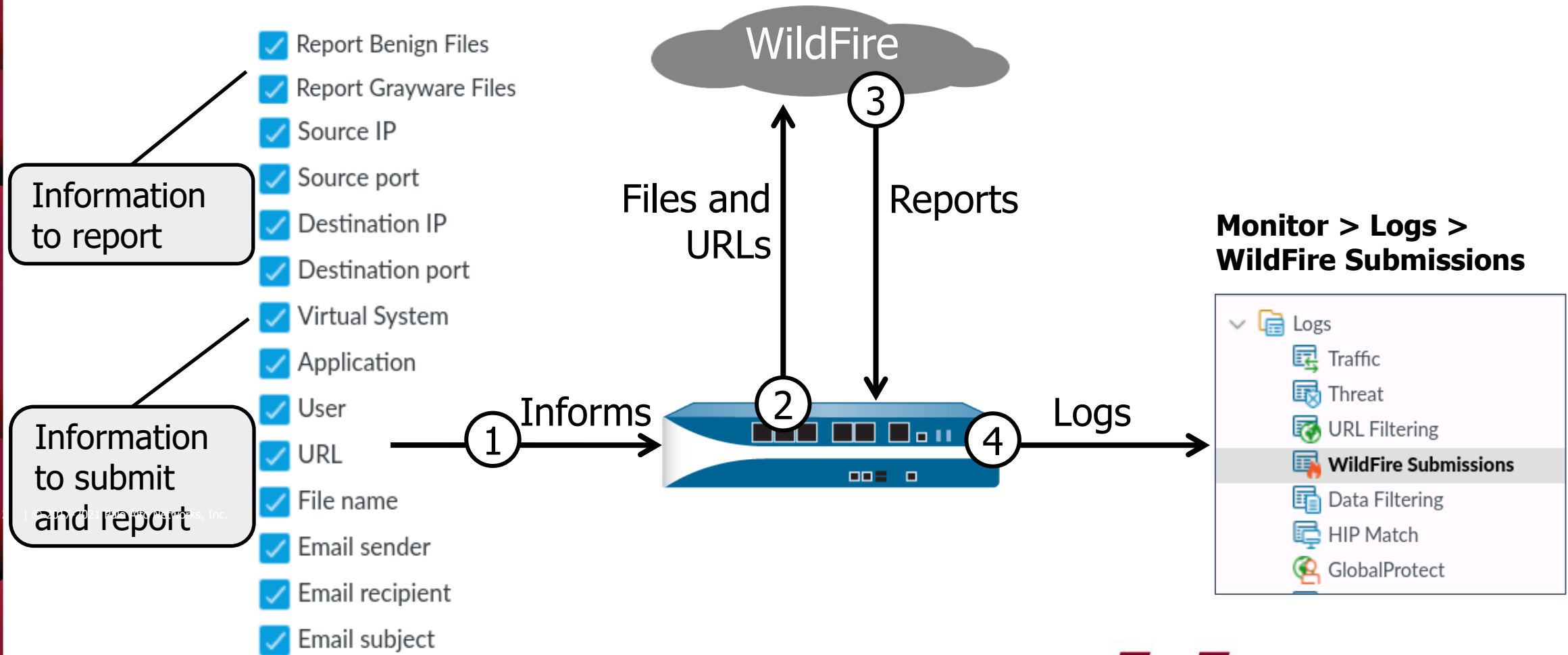
WildFire concepts

Configure and manage WildFire

WildFire reporting



WildFire Reporting



Verify Submissions and View Reports

> debug wildfire upload-log show

```
admin@firewall-a> debug wildfire upload-log show

Upload Log disk log rotation size: 2.000 MB.
Public Cloud upload logs:

log: 0, filename: wildfire-test-pe-file.exe
processed 6393 seconds ago, action: upload success
vsys_id: 1, session_id: 196, transaction_id: 3
file_len: 55296, flag: 0x801c, file type: pe
threat id: 52020, user_id: 0, app_id: 109
from 192.168.1.20/50731 to 52.20.176.145/80
SHA256: d6fbefe577a5336641f184ef4a3136889fed8fd0a37741165f01cd202549b637
```

CLI command to verify
successful file upload

View returned report
information.

Monitor > Logs > WildFire Submissions

	RECEIVE TIME	FILE NAME	SOURCE ZONE	SOURCE ADDRESS	APPLICATION	RULE	VERDICT	SEVERITY
	07/10 17:04:24	wildfire-test-pe-file.exe	Users_Net	192.168.1.20	web-browsing	Users_to_Internet	malicious	high

For a Tech Doc about this topic, log into Live and search
for "Verify WildFire Submissions"

WildFire Analysis Verdict Example

Monitor > Logs > WildFire Submissions

Detailed Log View

Log Info | **WildFire Analysis Report**

WildFire Analysis Summary [Download PDF](#)

File Information

File Type	PE
File Signer	
SHA-256	cc5ab063c9b9b05e830caed59b52097a43f66030572b13bb8a2bc55c7631d629
SHA1	1ee8da7a08a348288ad198e7cd2e8209af4997b3
MD5	a386b3ec09124906c0303f9d3f3ef4c1
File Size	55296 bytes
First Seen Timestamp	2020-07-22 17:22:11 UTC
Verdict	malware
Sample File	Download File

Download a PDF version of the report.

Download a copy of the file.

PCAP	RECEIVE TIME	TYPE	APPLICA...	ACTION	RULE	RULE UUID	BY...	SEVERI...	CATE...	URL CATE... LIST	VERDICT	URL	FILE NAME
	2020/07/22 17:22:40	end	web-browsing	allow	Users_...	542e5...	62...		any				
	2020/07/22 18:53:40	wildfire	web-browsing	allow	Users_...	542e5...		high			malicious		wildfir...

Close

Report Incorrect Verdict: Web Interface

Report Incorrect Verdict

Are you sure you want to report this file as having been incorrectly categorized as **malware**?

This session will be flagged for further analysis by Palo Alto Networks. When analysis is complete, you will be emailed with the results of the analysis and if necessary, the verdict in this report will be updated.

Sample Information

SHA-256	cc5ab063c9b9b05e830caed59b52097a43f66030572b13bb8a
MD5	a386b3ec09124906c0303f9d3f3ef4c1
Verdict	malware

Additional Information

Suggested verdict:

Your email address:

Future correspondence related to this report will be sent to this email address.

Please include any comments that may help us understand this issue more quickly.

- You can submit verdict change requests to Palo Alto Networks:
 - From web interface or WildFire portal
- From web interface:
 1. Select **Monitor > Logs > WildFire Submissions**.
 2. Find entry and click its detailed view icon.
 3. Click **WildFire Analysis Report** tab.
 4. Click **Select Incorrect Verdict** link.
 5. Suggest new verdict.

WildFire Portal

<https://wildfire.paloaltonetworks.com>



[Dashboard](#) [Reports](#) [Upload Sample](#) [Settings](#) [Account](#)

DASHBOARD

PREVIOUS 1 HOUR

Malware vs. Benign vs. Grayware vs. Phishing



■ Malware ■ Benign ■ Grayware ■ Phishing

Source					
Manual	13	24054	0	0	
013201004003	0	1	0	0	2022-11-15 10:26:07
007051000190754	0	1	0	0	2022-11-19 04:31:13
016401006387	0	29	0	0	2022-11-15 13:59:43
012801146927	0	1	0	0	2022-11-20 06:12:20

Click the firewall serial number to view list of submissions.

WildFire Dashboard Reports



DashboardReports

REPORTS

Search by file name or hash

- URL search doesn't support wildcards
- To make sure you receive results, use the file name or hash example, searching for 44c962009267ca4006c82b73e
- Note: depending on how you search, you may receive multiple results



WILDFIRE ANALYSIS REPORT

Download as PDF

FILE INFORMATION

File Type	JAVA Class
File Signer	
SHA-256	44c962009267ca4006c82b73ebed6478c8b9a6a4d264436904c112f13af222d6
SHA-1	e44fab80be764c65f71a4ddd153c569a4f23003
MD5	bd3932e014535efcf56b67a1a113e2ed
File Size	2682 bytes
First Seen Timestamp	2022-11-21 10:57:52 PST
Sample File	Download File
Verdict	Malware

Received Time		File Hash	Source IP	Destination IP	File Name	Verdict	Category
	2022-11-21 10:58:34	011901015617	200.1.44.37	200.1.5.85	ralunzpurteuzqjyyl/ncxwVw.class	Benign	unknown
	2022-11-21 10:58:33	011901015617	200.1.5.85	200.1.5.85	ralunzpurteuzqjyyl/ncxwVw.class	Benign	unknown
	2022-11-21 10:58:33	011901015617	200.1.5.85	200.1.5.85	ralunzpurteuzqjyyl/LSJpX.class	Benign	unknown
	2022-11-21 10:57:54	011901015617	200.1.9.72	200.1.9.72	ukgpnprfhsgdbfwqbskmbpwvcl/yktjxcks.class	Malware	unknown
	2022-11-21 10:57:52	011901015617	200.1.37.220	200.1.37.220	ukgpnprfhsgdbfwqbskmbpwvcl/yktjxcks.class	Malware	unknown

Report Incorrect Verdict: WildFire Portal

REPORTS

Source **Any**

	Received Time	Source	File / URL
	2020-03-30 14:56:10	Manual	mega_feb2020 FB-651.pdf
	2020-03-30 14:56:10	Manual	
	2020-03-30 14:56:10	Manual	

REPORT INCORRECT VERDICT

Are you sure you want to report this file as being incorrectly categorized as **benign**?

This sample will be flagged for further analysis by Palo Alto Networks. If you chose to give us your email address, you will receive an email with the results of the analysis review. If the verdict is found to be incorrect, it will be updated to the correct verdict in all previous and future WildFire logs featuring this sample.

SAMPLE INFORMATION

SHA-256	458f0825f25b10f4fefa6255ea473f3ca8416cc0a10da73326d84077f29293f8
MD5	4bf9885dff08be26c5a7aa73a005a26d
Verdict	Benign

ADDITIONAL INFORMATION

Suggested verdict: **Malware**

Email:

Future correspondence related to this incorrect verdict report will be sent to the email address

REPORT TO PALO ALTO NETWORKS

This sample was determined to be benign. If you believe this verdict is incorrect, please [report an incorrect verdict](#). This action will send sample to Palo Alto Networks for further analysis.

Module Summary

Now that you have completed this module, you should be able to:

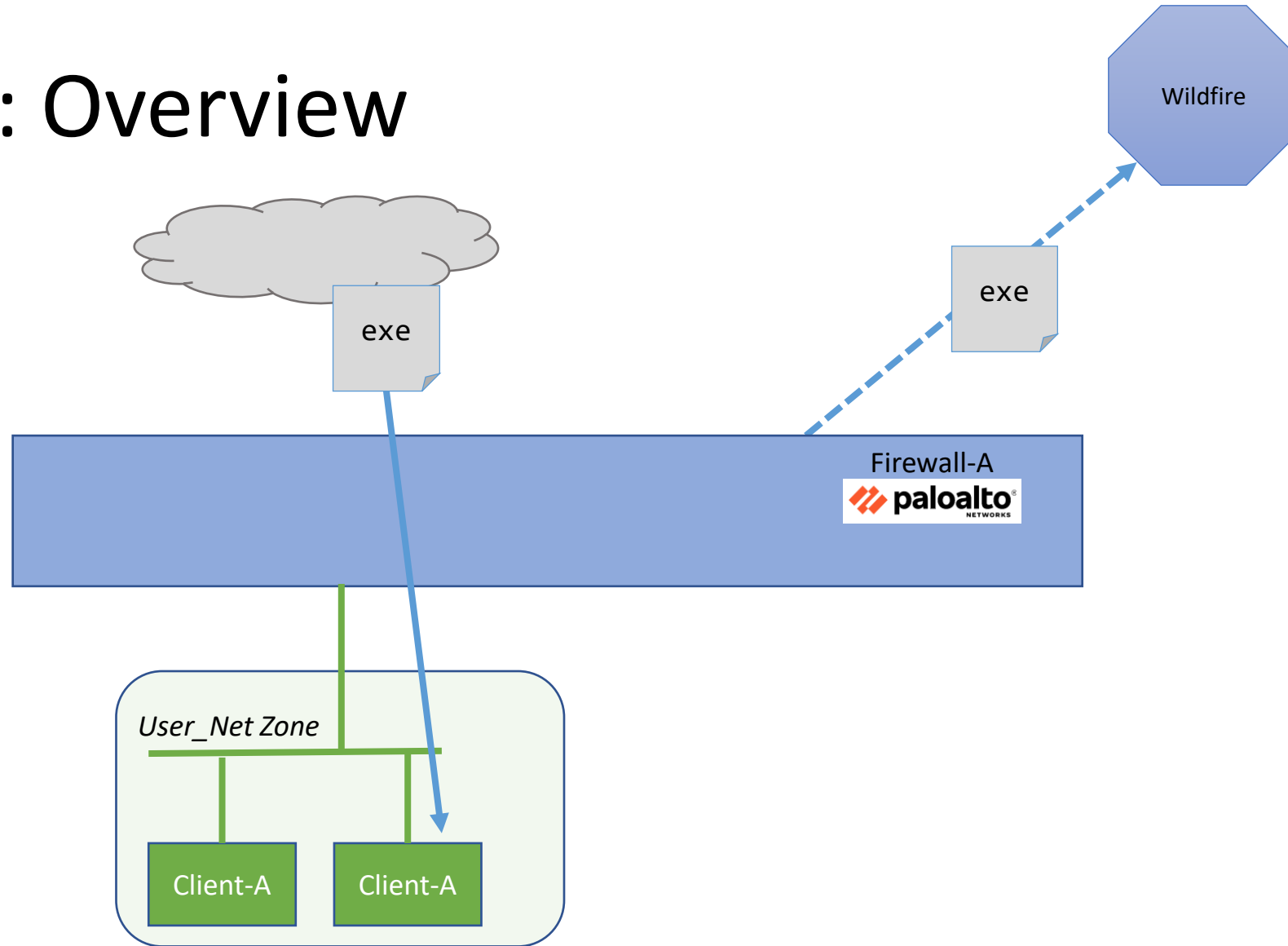
- Describe WildFire purposes and operation
- Describe WildFire license and deployment choices
- Configure and update WildFire
- View WildFire reports and logs



Questions



Lab 10: Overview



Lab 10: Blocking Unknown Threats with WildFire

- Create a WildFire Analysis Profile
- Apply WildFire Profile to security rules
- Test the WildFire Analysis Profile
- Examine WildFire analysis details

End Content

Unit 11 Homework

- Always check Blackboard for assignments and when they are due.
- Unit 11 Lab: Netlabs Lab 10 – Blocking Inappropriate Web Traffic with Advanced URL Filtering
- Unit 11 Assignment - ML-Based Prevention Video
- Unit 11 Quiz